MIL-P-38005B(USAF)
4 November 1965
Superseding
MIL-P-38005A(USAF)
9 October 1964

MILITARY SPECIFICATION

PITOT-STATIC TUBE

1. SCOPE

1.1 This specification covers the requirements for aerodynamically compensated pitot-static tubes.

1.2 Classification. Pitot-static tubes shall be classified as follows:

MS27376-1	Right-hand
MS27376-2	Left-hand

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5509	Fittings, Fluid Connection
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence For the Selection of
MS3106	Connector, Plug, Electric, Straight
MS3107	Connector, Plug, Electric, Quick Disconnect
MS3108	Connector, Plug, Electric, Angle 90 Degree
MS27376	Pitot-Static Tube, 115 Volt, 400 Cycle, Fuselage Mounted
MS33586	Metals, Definition of Dissimilar
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The tube furnished under this specification shall be a product which has been tested and has passed the qualification tests herein,

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and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials which are not specifically designated shall be of the best commercial quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the tube except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used in all parts of the tube except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction

3.4.1 Design. The design of the tube shall conform to figure 1 and MS27376 and shall be such that the tube will sense total and static pressures within the accuracies specified herein. Aerodynamic compensation shall be achieved by appropriate contouring of the pressure sensing portion (sections A and B) of the tube.

3.4.2 Construction. The tube shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

*3.5 Performance. The tube shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Insulation: Potential voltage of 700V rms at sea level

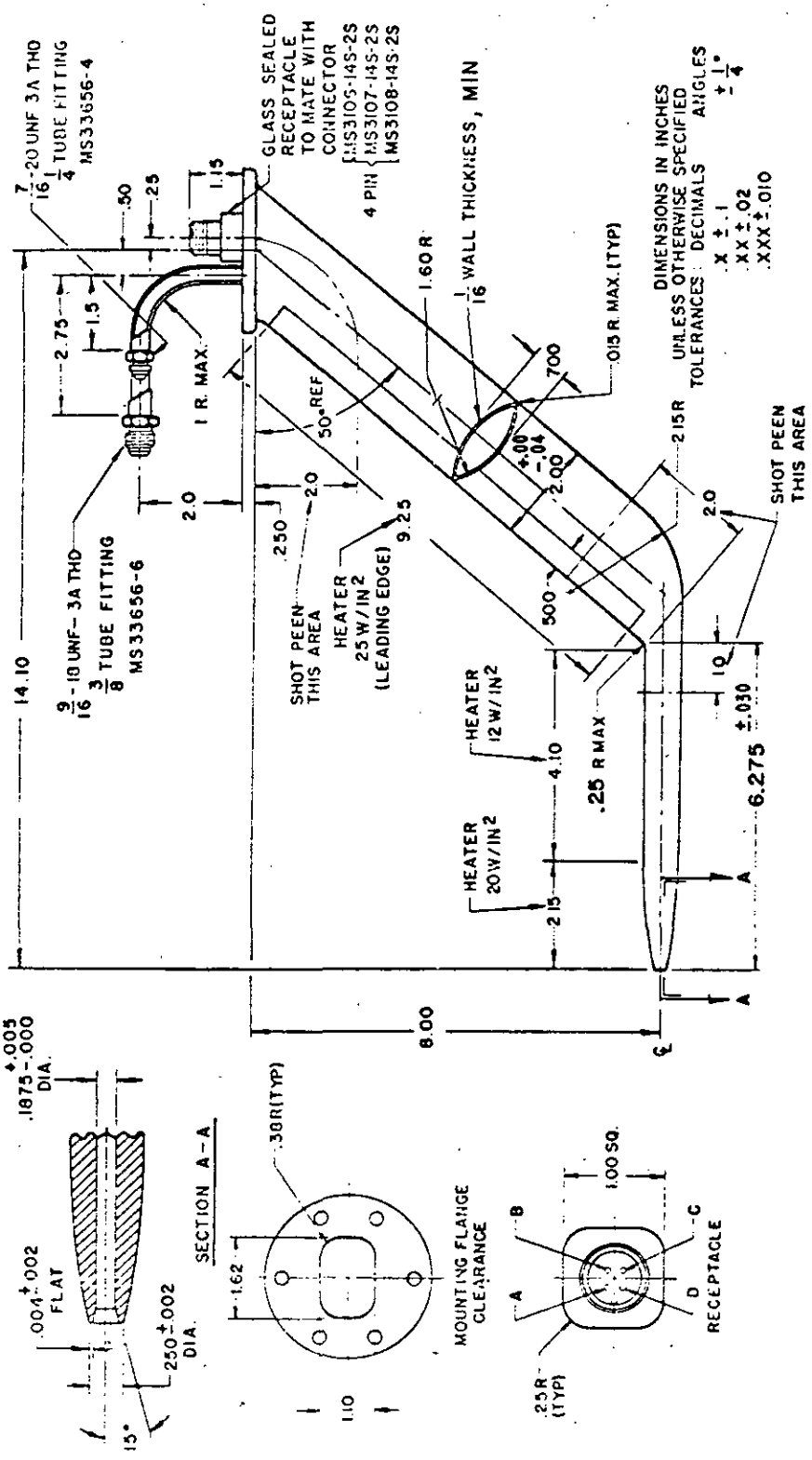
b. Pressure leakage: Pressure of 80 inches Hg for a period of 1 minute

c. Vibration: Vibration for 2 hours including resonant and cycling conditions in each of the horizontal and vertical axes

d. Endurance cycling: Operation at rated voltage for a period of 10 hours, then off 1 hour for a total of 30 cycles

e. Deicing and anti-icing: Removal and complete clearance of accumulated ice for 15 minutes with the tube positioned at 0° and 25° angles of attack, when subjected to wind tunnel speeds of 350 knots at temperatures from -25° to -35°C with a liquid water content of 1.25 grams per cubic meter, and with both heaters operating. Removal and clearance of accumulated ice for 15 minutes under the same conditions except with only the head heater operating

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★ FIGURE 1. Pitot-Static Tube Configuration

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f. Static and total pressure error: Wind tunnel tests at mach numbers ranging from 0.2 to 0.95 throughout an angle of attack ranging from -4° to +24° in 2° increments

g. Salt spray: Exposure to salt sea atmosphere for a period of 48 hours without damage

h. High temperature: Exposure to 400°C for a period of 48 hours without damage

i. Low temperature and power consumption: Exposure to -65°C for a period of 48 hours without damage; after 24 hours at -65°C, shall withstand an initial power surge not exceeding 1,325w with both heaters operating and 525w with only the head heater operating

j. Humidity: Relative humidity up to 100 percent.

3.5.1 Power consumption. After 5 minutes of operation at rated voltage and under still-air conditions, electrical power consumption shall not exceed 250 ±25w with both heaters operating nor 105 ±20w with only the head heater operating.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.7 Electrical circuits. The tube shall have two 2-wire type electrical circuits as shown on figure 2, one for the head heater and one for the mast heater. The electrical connector shall be Sealtron part No. 8001-14S-2P, or equivalent.

3.8 Heaters. The tube shall have two heaters, one located in the head of the tube and one in the mast. During the deicing operation, power distribution shall be in accordance with the minimum wattage densities shown on figure 1, with the heater in the head corresponding to heater sections A and B, and the mast heater corresponding to heater section C. The total overload wattage with both heaters operating shall not exceed 1,325w and with only the head heater operating shall not exceed 525w. Heater surfaces shall be placed as close as practicable to the surfaces to be heated. The heaters shall be regulated automatically in such a manner that the power dissipated through the heaters will be an inverse function of the heater-element temperature. In still air at an ambient temperature between 20° and 30°C, the combined output of both heaters shall not exceed 250 ±25w after 5 minutes of operation and for the head heater alone shall not exceed 105 ±20w.

3.9 Heat distortion. There shall be no permanent distortion of the tube body as a result of extended heater operation.

3.10 Pressure lines. The total pressure line shall consist of 1/4-inch outside diameter tubing with a wall thickness of 0.022 inch. The line fitting shall conform to MIL-F-5509 and MS33656-4. The static pressure line shall consist of 3/8-inch outside diameter tubing with a wall thickness of 0.035. The static pressure line fitting shall conform to MIL-F-5509 and MS33656-6.

3.11 Total pressure chamber. A plenum chamber shall be provided, the volume of which shall not exceed 1.75 cubic inches. The pressure lines shall be so connected to the plenum chamber that no moisture can enter the lines.

3.12 Static pressure chamber. The internal volume of the static pressure chamber shall be no less than the internal volume of 3/8-inch outer diameter tubing (0.035 inch wall thickness) having the same length as the static pressure chamber.

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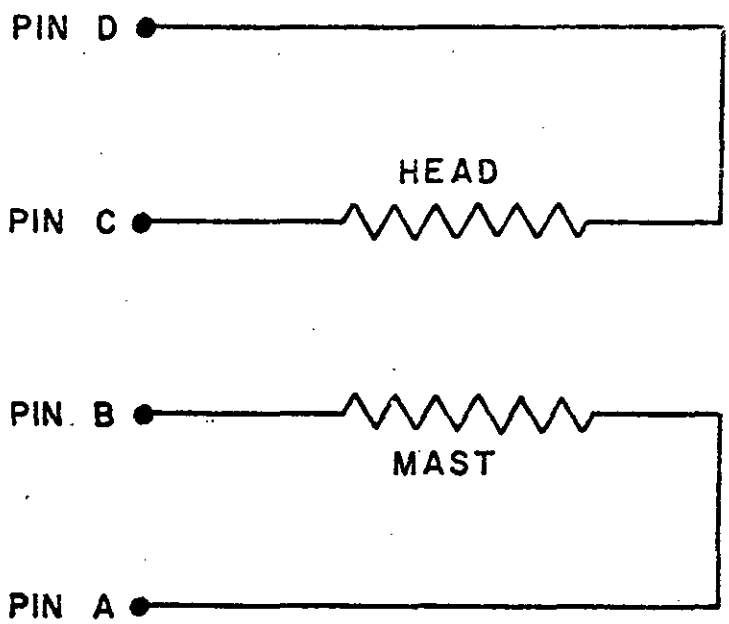


FIGURE 2. Wiring Diagram

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*3.13 Concentricity. The compensating nose contour and the cylindrical portion of the tube body shall be concentric within ±0.001 inch. There shall be no surface irregularities in the vicinity of the static pressure orifices.

3.14 Drainage. The tube shall be so designed that water will not enter the pressure line under any service condition. A drain hole shall be provided in the total pressure chamber only, to clear any trapped moisture. The diameter of the drain hole shall be $0.035\ ^{+0.000}_{-0.005}$ inch.

*3.15 Screw threads. The threads of all machine screws shall conform to MIL-S-7742.

*3.16 Weight. The tube shall be weighed without the specified electrical connector and shall weigh not more than 2.5 pounds.

3.17 Finish. Protective coatings and finishes that will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used. The outside of the tube shall have a durable dull finish acceptable to the procuring activity.

3.18 Protective caps. The total pressure opening, electrical connector, and pressure line coupling threads shall be protected by slip-on caps of nonstrategic material. The design of all protective devices shall be subject to the approval of the procuring activity.

3.19 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Identifying information shall be etched, engraved, or stamped on each tube in a location that will not interfere with airflow over the tube or with tube operation.

*3.19.1 Other markings. The total pressure and static lines shall be identified by the letters P and S, respectively, and shall be stamped, etched, or engraved on the coupling.

3.20 Workmanship. The tube shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of tubes shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified, the test shall be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above

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values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Rated voltage. Unless otherwise specified, the tube shall be operated on a rated voltage of 115V, 400-cycle ac.

4.3.3 All tests shall be conducted with the mating electrical connector installed.

4.4 Qualification testing

*4.4.1 Test samples. The test samples shall consist of three tubes representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The contractor shall submit the samples that were tested to the procuring activity.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests described under 4.6.

*4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each tube shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Insulation

c. Pressure leakage

d. Power consumption.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One tube shall be selected at random from each 40 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Vibration tests.

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 tubes shall be selected at random from the first 15 produced on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Static and total pressure error

c. Deicing and anti-icing

d. Salt spray

e. High temperature

f. Low temperature and power consumption

g. Endurance cycling

h. Humidity.

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test

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failure could indicate that defects may exist in items already accepted. If so, the contractor shall advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The tube shall be carefully examined to determine compliance with the requirements specified herein with respect to materials, finish, weight, markings, and workmanship.

*4.6.2 Insulation. The tube shall be subjected to a high-potential test at sea level pressure with 700V rms commercial frequency applied between each terminal and the case for a period of 1 minute. Upon completion of testing, there shall be no damage to the wiring or insulation. In the event a hermetically sealed heater is utilized, the tube shall be placed in a chamber and evacuated to a pressure equal to an altitude of 80,000 feet for a period of 4 hours. At the end of this period, 150V ac shall be applied between each terminal and the case for 60 seconds. No damage to the wiring or insulation shall result.

4.6.3 Pressure leakage. A pressure of 80 inches Hg shall be applied to each of the pressure lines. With the total and static pressure orifices and drain holes sealed, the applied pressure after 1 minute shall be not less than 79.97 inches Hg.

4.6.4 Power consumption. At an ambient temperature between 20° and 30°C, rated voltage shall be applied through a fuse to the terminals of the tube for operation of both heaters for a period of 5 minutes. After the 5-minute period, the power shall be measured and shall not exceed 250 ±25w.

4.6.4.1 At the conclusion of the test specified in 4.6.4, rated voltage shall be applied through a 3-amp fuse to the head heater for a period of 5 minutes. At the end of this period, the power shall be measured and shall not exceed 105 ±20w.

4.6.5 Vibration. The tube shall be mounted in a sleeve at the aft end and subjected to vertical and horizontal vibrations perpendicular to the tube centerline in accordance with procedure XII of MIL-E-5272. Rated voltage shall be applied to both heaters during the entire test. No leaks or failures shall result from this test.

4.6.6 Deicing and anti-icing. The tube shall be tested in an icing wind tunnel at an indicated tunnel speed of 350 ±25 knots. The ambient temperature for each test shall be -30° ±5°C. The water droplet size shall be regulated to provide a liquid water content of 1.25 ±0.25 gram per cubic meter. Test runs shall be made at angles of attack of 0° and 25°. The procedure for each test run shall be as follows: Ice shall be allowed to form on the tube until either (a) the total pressure opening has been sealed or (b) the icecap has extended 1/2 inch in front of the tube tip. Power shall then be applied to both heaters at rated voltage and the total time required to achieve correct pressure readings shall not exceed 1 minute; the time required to remove all accumulated ice shall not exceed 1 minute and 30 seconds. The icing test shall be continued for 15 minutes after the icecap is removed and no ice shall collect on the tube that will affect the pressure indication.

4.6.6.1 The test specified in 4.6.6 shall be repeated except that after icing occurs as specified, power at rated voltage shall be applied to the head heater only. The time required to achieve correct pitot pressure shall not exceed 1 minute, and the time required to remove all accumulated ice from heater sections A and

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B shall not exceed 1-1/2 minutes. This test shall be continued for 15 minutes after the icecap has been removed; no ice shall collect on the tip of the tube or on heater sections A and B.

* 4.6.7 Static and total pressure errors. The tube shall be mounted in a wind tunnel and tested to determine static and total pressure errors. The tube errors shall be determined by comparing the pressures sensed by the tube with actual tunnel pressures. The tube shall be tested at speeds from 0.2 to mach 0.9 inclusive in increments of 0.1 mach, and at mach 0.95. At each test point 0.6 mach or greater, the tube shall be tested at angles of attack from -4° to +14°; at test points less than 0.6 mach, the tube shall be tested at angles of attack from -4° to +24°. In both cases, the increment of test shall be 2°. The pressures sensed by the static ports shall be such that when the tube is installed on the aircraft in the location specified (see 6.3), the static pressure errors at that location will be corrected so that the residual error will not exceed the tolerances shown on figure 3. At angles of attack other than 0°, the accuracies shall be as shown on figure 4. The total pressure sensed by the tube at 0° angle of attack shall not differ by more than 1/2 percent from the actual total pressure in the tunnel. At other angles of attack, the total pressure sensed by the tube shall not differ by more than 1 percent from the actual total pressure in the tunnel.

4.6.8 Salt spray. The tube shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272.

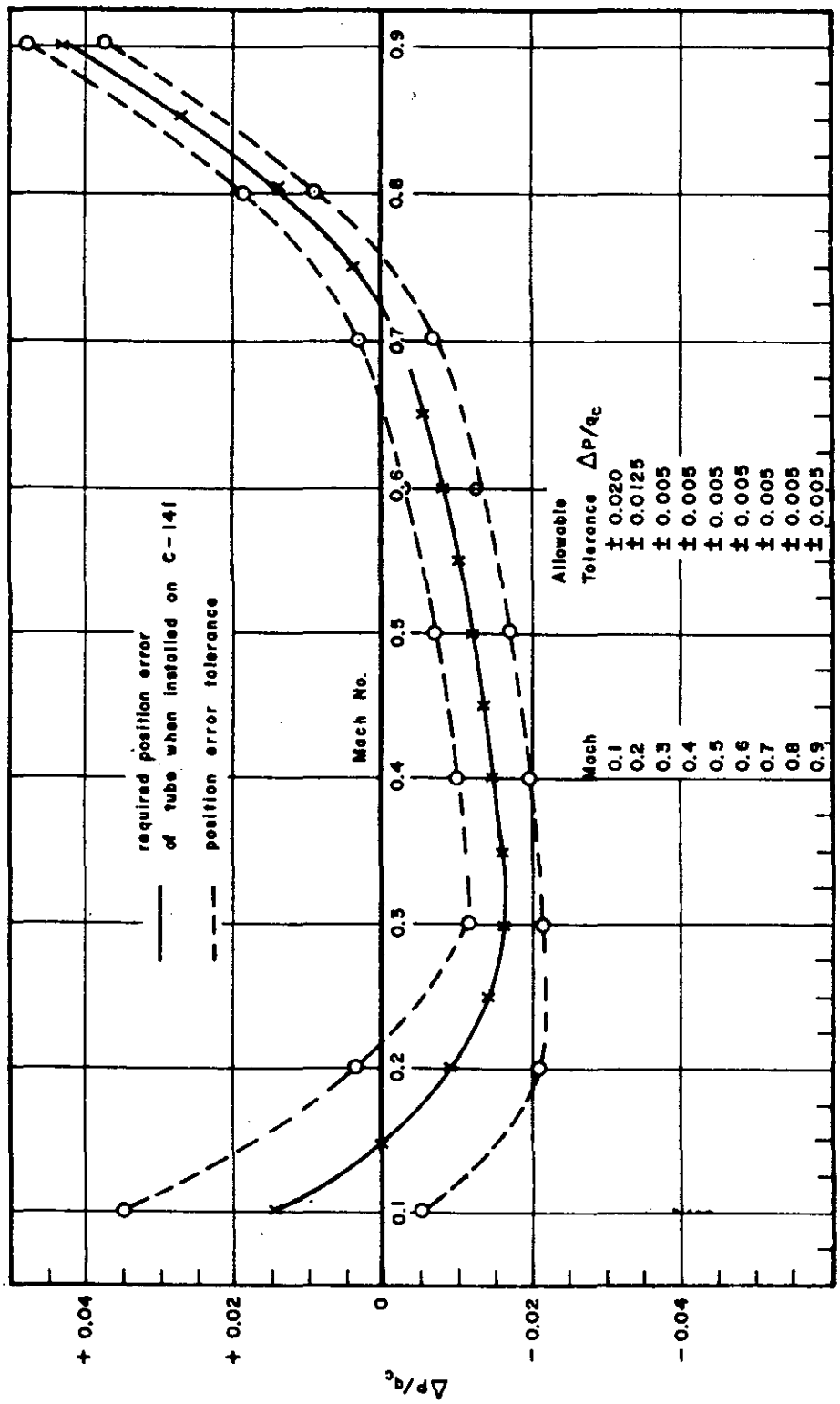
4.6.9 High temperature. The tube shall be subjected to a temperature of 400°C for a period of 48 hours. At the end of this time, there shall be no evidence of damage. At the option of the procuring activity, any or all tests may be conducted after this test.

4.6.10 Low temperature and power consumption. The tube shall be subjected to a temperature of -65°C for a period of 48 hours. After 24 hours at low temperature, rated voltage shall be applied to the tube through a 10-amp circuit breaker for operation of both heaters. The initial power surge shall not exceed 1,325w and shall not be of such a duration as to render the circuit breaker inoperative. After 5 seconds, the power consumption shall not exceed 775w and after 20 seconds shall not exceed 550w. Power shall be removed following the power consumption test. At the option of the procuring activity, any or all of the tests may be conducted after this test.

4.6.10.1 The test specified in 4.6.10 shall be repeated except that after 24 hours at low temperature, rated voltage shall be applied through a 10-amp circuit breaker to the head heater only. The initial power surge shall not exceed 525w and shall not be of such duration as to render the circuit breaker inoperative. After 5 seconds, the power consumption shall not exceed 300w and after 20 seconds shall not exceed 225w. If the test is conducted immediately after the test specified in 4.6.10, the low temperature soak period may be reduced to 6 hours.

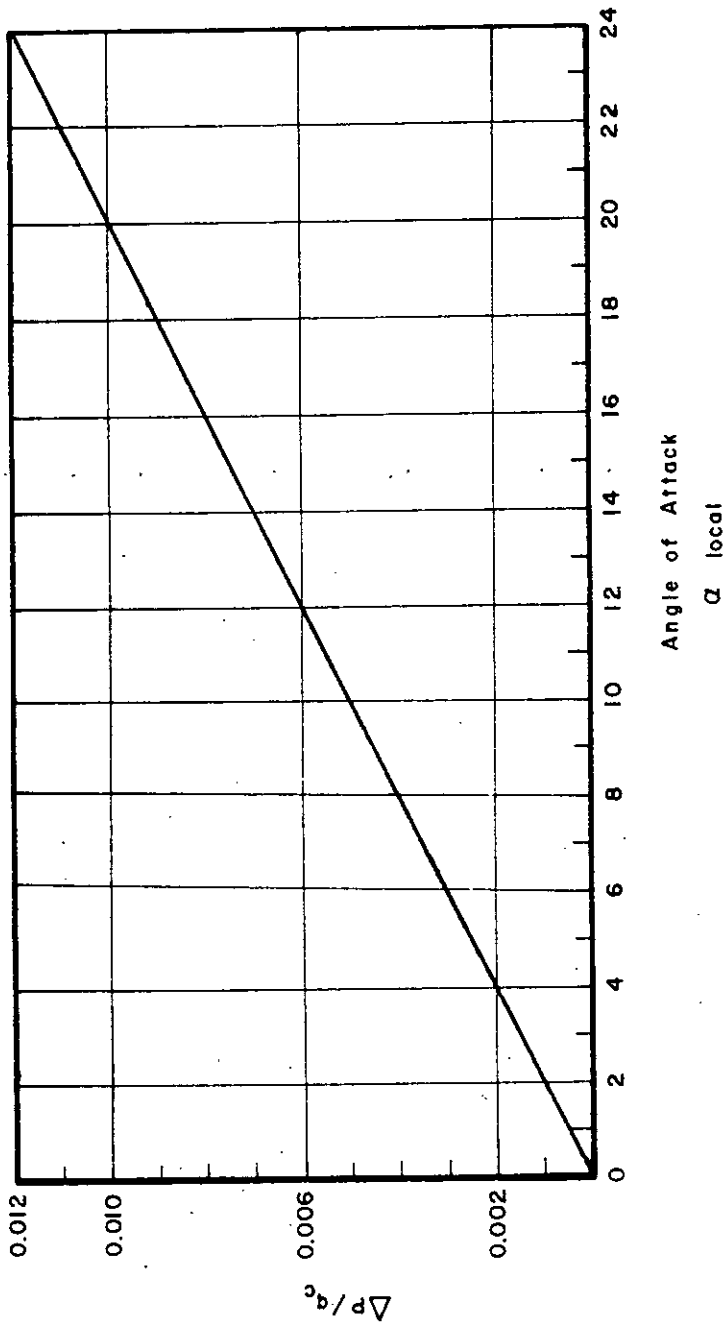
4.6.11 Endurance cycling. At an ambient temperature of between 20° and 30°C, the tube shall undergo an endurance cycling test in which rated voltage is applied to both heaters simultaneously for an uninterrupted period of 10 hours and then removed for 1 hour. The test shall be repeated for 30 consecutive cycles. As a result of this test, there shall be no damage which will affect corrosion resistance or proper operation of the tube. At the conclusion of this test, the tube body shall be checked dimensionally to determine that no permanent

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★ FIGURE 3. Residual Error Tolerances

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★ FIGURE 4. Additional Tolerances for Static Pressure Sensing at Angle of Attack

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distortion or warpage has occurred due to heater operation. The pressure leakage and insulation tests shall then be repeated after which the tube body shall again be checked dimensionally to determine that no permanent distortion or warpage has resulted due to heater operation.

4.6.12 Humidity. The tube shall be subjected to a humidity test in accordance with procedure I of MIL-E-5272. There shall be no detrimental effect on the operation of the heaters nor deterioration of the tube body. At the completion of this test, the tube shall meet the test specified in 4.6.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each pitot-static tube shall be preserved in accordance with MIL-P-116, method 1A8 and individually packaged in accordance with MIL-P-7936.

5.2 Packing

5.2.1 All levels. Each pitot-static tube shall be packed in accordance with MIL-P-7936. Containers for domestic or overseas shipment shall conform to PPP-B-636.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tube covered by this specification is intended for use on C-141 aircraft to supply total and static pressures for flight instruments and control equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing.

6.3 Tube location. The C-141 aircraft pitot-static tube static port location is at FS 518, WL 243. Figures 5 and 6 give the pressure field vs angle of attack and mach number for that location. Based upon consideration of the figure 5 data, a suitable contour and location of the static ports should be determined that will provide the performance specified herein.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to the requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group, Attn: SEFIF, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

*6.4.1 Test report. The contractor will submit to the procuring activity three copies of a test report prepared in accordance with MIL-STD-831.

6.5 The outside margins of this specification have been marked to indicate where changes from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any

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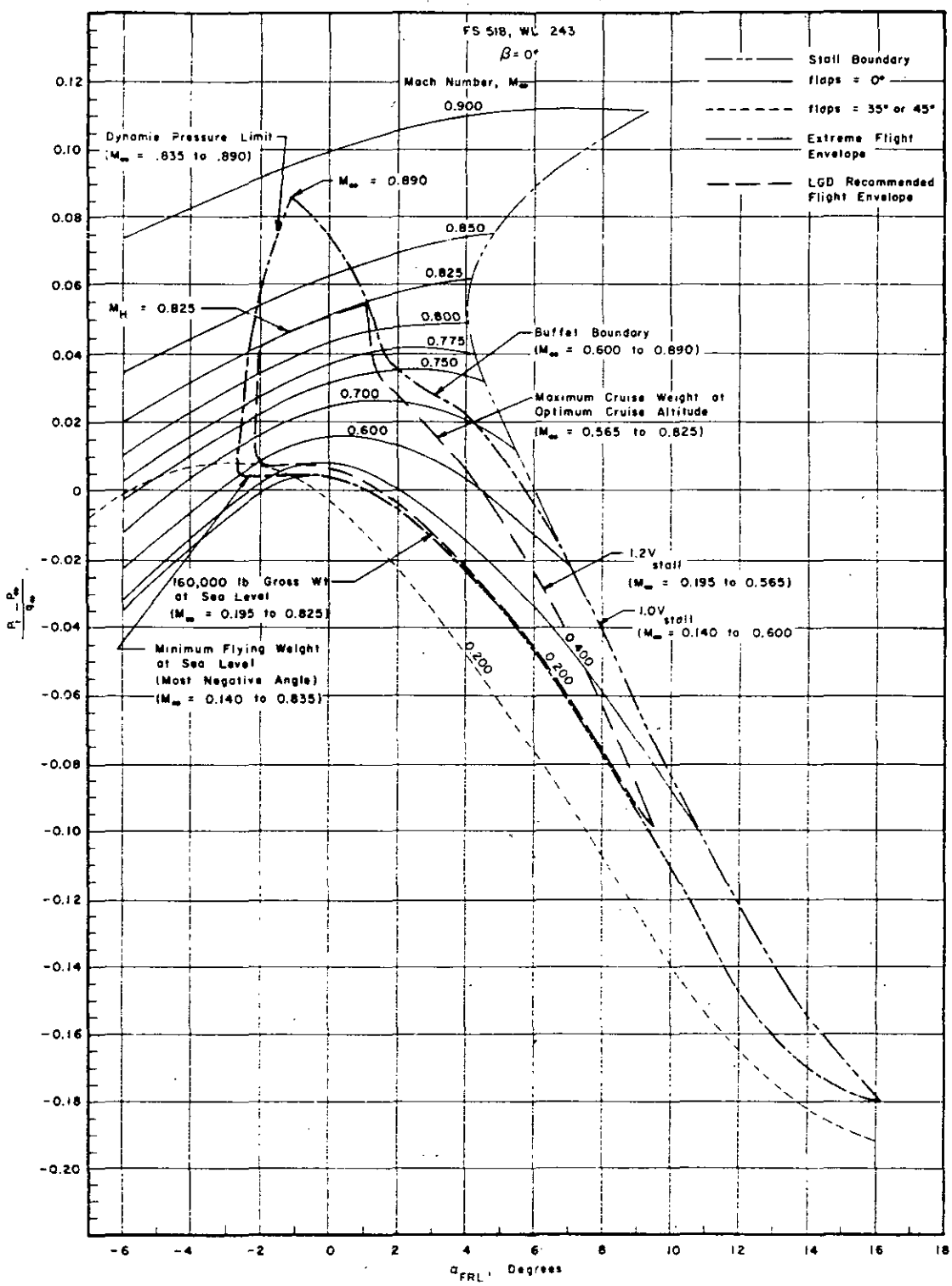


Figure 5. Pressure Coefficient at Pitot-Static Tube Ports Location

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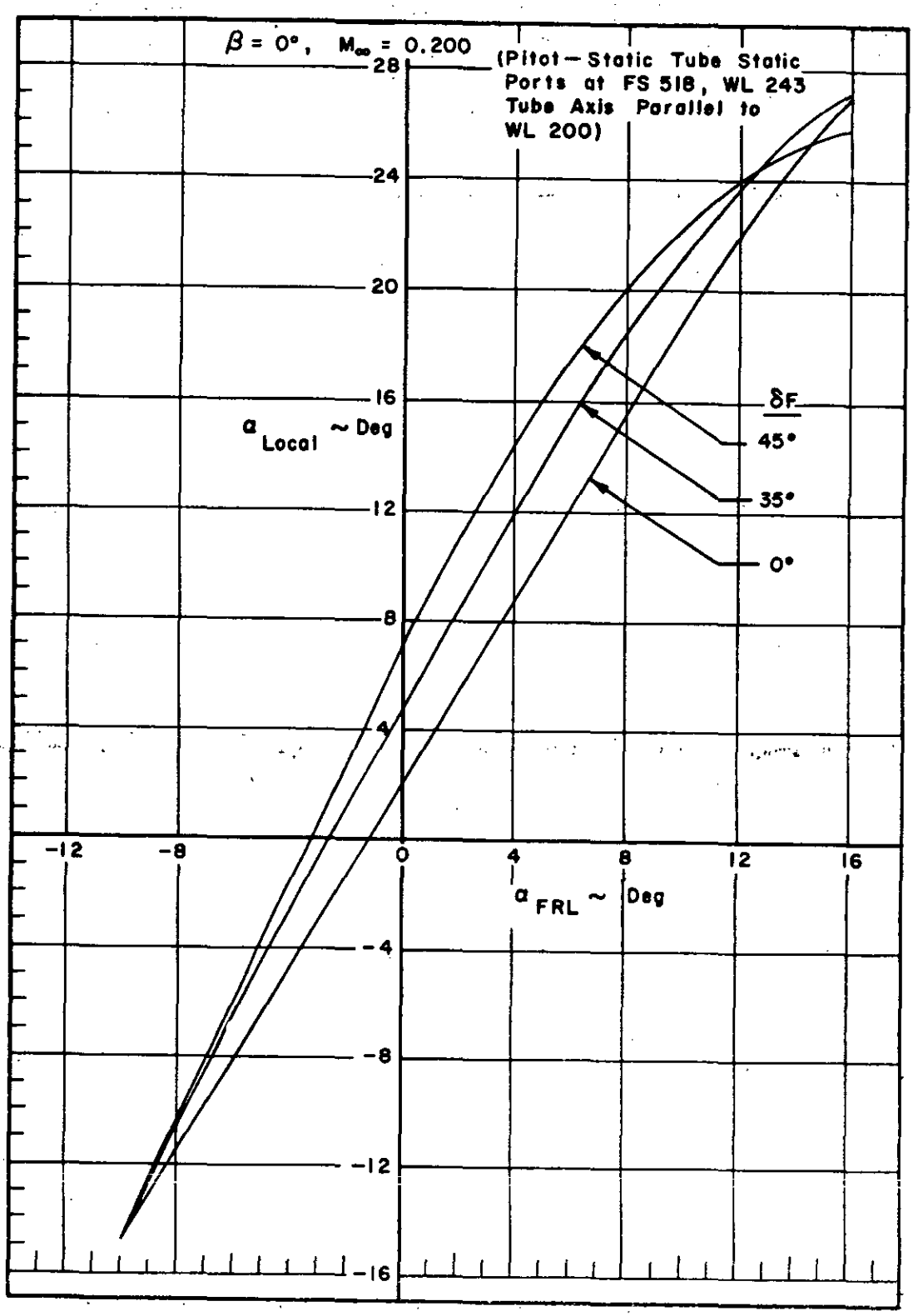


FIGURE 6. Local Angle of Attack of Pitot-Static Tube Vs α FRL

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inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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